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RESEARCH ARTICLE

Iterative Assessment of Edge Criticality: Efficiency Enhancement or Hidden Insufficiency Detection

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ABSTRACT The assessment of edge criticality ranking in complex networks is a challenging issue in network science and has numerous applications, including network decomposition and, conversely, enhancing the resilience and redundancy of complex systems. Two main approaches are commonly used to rank edges based on their importance for maintaining network connectivity. The first is the Static approach, which relies on a single evaluation of topological features. The second is the Optimization-based approach, which treats network decomposition as an integral process and optimizes the edge sequence for network decomposition using genetic-like algorithms. While the Static approach is computationally efficient, the Optimization-based approach potentially yields the best decomposition pattern. In the present work, we propose the Iterative approach, which bridges the gap between these two methods. The Iterative approach involves a loop of identifying the most critical edge using selected ranking algorithms, removing it from the network, and then re-assessing edge criticality based on the modified network topology. As a result, the ranking of edge criticality depends not only on the initial topology of the network but also on its continuous modifications caused by edge removal. To evaluate the efficiency of the Iterative approach, we analyze the decomposition of sixteen well-known real-world benchmark networks using seven widely recognized edge ranking algorithms. The results demonstrate, first, that the Iterative approach can achieve a tenfold increase in the efficiency of network decomposition. Second, the analysis reveals hidden inner insufficiency in edge ranking for some algorithms, as evidenced by the fact that algorithm iterations can reduce decomposition efficiency. Additionally, we discuss the time complexity of the Iterative approach and strategies for its reduction. We also outline a potential framework for combining the Static and Iterative approaches during the network decomposition process to further enhance its efficiency.

INDEX TERMS Bridgeness, complex networks, critical edges, degree product, diffusion importance, edge betweenness, iterative approach, k -path centrality, link entropy, topological overlap.

I. INTRODUCTION

The identification of critical elements in complex networks is crucial for a variety of practical applications [1], including ensuring the smooth operation of many complex networks in modern society [2], [3], [4]. In particular, in social networks, critical nodes facilitate the flow of key information [5], [6], [7], while in supply chain networks, they are vital for

ensuring network robustness and resilience [8], [9], [10] and preventing cascading failures [11], [12]. Critical elements also play a pivotal role in controlling the performance of urban transportation infrastructures and communication systems [13], [14], [15]. Furthermore, they are instrumental in research on disease spread and vaccination strategies [16], [17], [18].

Related directions in modern network research delve into the development and optimization of networks. These include the prediction and planning of new links and future

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communication channels [19], analytical support for medical drug development [20], and the analysis of multiplex and neural networks [21], [22].

In the present work, we focus on the critical edges (links) of complex networks and examine their robustness against partial destruction. Within this context, identifying links that are critical for preserving network connectivity represents a key challenge. It has been discussed in the context of maintaining global connectivity [23], intentional attacks or unpredictable failures of power systems and communication systems [24], disseminate information [25], online political communications [26], [29], controlling the flow of information and maintaining the integrity of transportation, communication, and social networks [27], effective decomposition of harmful networks, such as terrorist networks, rumor spreading networks, infectious disease spreading [28], common aspects of networks related to attack vulnerability analysis, the spread of epidemic diseases, and information spreading patterns discovery [30]. Appropriate quantification of edge significance and the associated modification of network topology can substantially enhance the resilience and robustness of networks against targeted edge attacks [31]. This is because the loss of even a small number of critical links in a network can trigger cascading failures, ultimately leading to system collapse [14], [32], [33]. Another aspect of this problem involves identifying the minimal set of edges whose removal can effectively decompose a network. For instance, using data from South Korea, it was demonstrated that removing key links in social networks can mitigate the spread of contagious diseases by significantly reducing the size of the infection network [34].

The aforementioned systems define the domain of the given problem's applications. The robustness of such networks critically depends on identifying key edges and understanding possible actions, including modifications to the network topology, aimed at preserving network integrity. It should be emphasized that network modification is not a purely local problem, as reinforcing currently critical edges may cause other edges to become "weak links." In particular, this dynamic aspect is central to the proposed Iterative approach.

A. RELATED WORKS

Nowadays, a number of prominent algorithms have been developed for quantifying edge significance [23], including (see also Sec. III-C):

- *Edge Betweenness Centrality* by Girvan and Newman [35], which quantifies edge significance based on the collection of shortest paths—a key factor for efficient communication;
- *Degree Product* introduced by Wang and Chen [36] for weighted networks, focusing on edges connected to high-degree nodes;
- *Bridgeness* by Cheng et al. [37], determining the uniqueness of edges serving as bridges;

- *Diffusion Importance* constructed by Liu et al. [38], from the perspective of spreading dynamics;
- *Topological Overlap* by Onnela et al. [13], evaluating the overlap of neighboring nodes;
- *Link Entropy* by Qian et al. [23], quantifying edge significance from the perspective of community distribution;
- *k-path Centrality* by Alahakoon et al. [39] and Meo et al. [40]—an approach to quantifying edge significance in social networks, similar to Edge Betweenness, but with the condition on path shortness eliminated.

Network decomposition is typically analyzed using the listed algorithms, among others, to quantify edge significance at the first step only. In other words, these algorithms rank the edges of the initial network according to their significance, and the subsequent network decomposition is performed by removing the edges in the determined order. We refer to this procedure as the *Static approach* to network decomposition. In contrast, within the *Evolutionary approach* [30], which employs a genetic-like algorithm, network decomposition is determined by minimizing the area under the curve that represents the dependence of the largest connected component on the number of removed edges. As demonstrated, the Evolutionary approach exhibits significantly higher efficiency compared to the Static approach. This can be attributed to the fact that edge removal inherently alters the network topology; thus, the initially constructed edge order eventually becomes suboptimal, leading to a loss of efficiency in further decomposition. Naturally, the drawback of the Evolutionary approach lies in its computational time cost.

The purpose of this paper is to develop a novel approach that partially bridges the gap between the Static and Evolutionary approaches. In other words, this approach is expected to be more efficient than the Static approach while being less time-consuming than the Evolutionary approach.

B. CONTRIBUTION

In this work, we aim to draw attention to the effect of re-arranging edges according to their significance during network decomposition. To this end, we adopt the *Iterative approach* for assessing edge criticality, applied to unweighted and undirected networks. As a preliminary result, we observe that re-evaluating edge significance using the Improved Link Entropy [29], followed by re-arranging edges, can enhance the efficiency of network decomposition. Similarly, in the context of community detection, which forms the basis for edge significance in the Link Entropy [23] and Deep Link Entropy [26] algorithms, changes caused by edge removal must be accounted for to further improve network decomposition.

As a specific contribution of this paper, it is worth noting that the analyzed network decomposition governed by the Iterative approach has revealed two distinct aspects:

- As expected, the Iterative approach can be more efficient than the Static approach. A novel result is that

for several benchmark networks, the Iterative approach achieves a tenfold increase in decomposition efficiency compared to the Static approach.

Unexpectedly, the Iterative approach can reveal certain hidden insufficiency of well-known algorithms when applied to network decomposition. Specifically, for some networks, its use results in a decrease in efficiency, prompting the question of which critical network parameters distinguish these cases from others—parameters that typically remain outside the scope of investigation.

To minimize potential biases, we apply the Iterative approach to decompose sixteen benchmark networks of varying sizes and origins, utilizing seven well-known algorithms: *Edge Betweenness*, *Degree Product*, *Bridgeness*, *Diffusion Importance*, *Topological Overlap*, *Link Entropy*, and *k-path Centrality*. Finally, the results are evaluated statistically.

The rest of the paper is organized as follows. Section II discusses the general criteria for network decomposition used in the numerical analysis of its progression. Section III presents the pseudocodes for the Static approach (Sec. III-A) and the Iterative approach (Sec. III-B) and summarizes the ranking indices of the main algorithms underlying network decomposition (Sec. III-C). Section IV describes the analyzed networks (Sec. IV-A) and presents the obtained results (Sec. IV-B). In Section V, these results are discussed in detail, and their implications are examined. Finally, Section VI summarizes the findings and provides concluding remarks.

II. CRITERION OF ALGORITHM EFFICIENCY

In comparing the efficiency of two algorithms for decomposing a single network, the size of the Largest Connected Component (LCC) as the network is destroyed often serves as a quantitative criterion [23], [25], [26], [41]. A smaller LCC indicates more significant edge removal, suggesting more efficient destruction of the network's connectivity. Therefore, if for two algorithms, A_1 and A_2 , the inequality $LCC_1 < LCC_2$ holds throughout most of the decomposition process, the first algorithm is considered more efficient.

However, the use of this criterion becomes problematic in two cases [23], [25], [42], [43]. First, when two algorithms exhibit similar efficiency, the order of their LCCs can change during the intermediate stages of network decomposition. Second, when comparing networks of significantly different sizes, referring to the absolute values of the LCC can be misleading.

To address these issues, we modify the LCC criterion. In the first step, we replace the size of the LCC with the Relative Size of the Largest Connected Component. Specifically, the quantity used for algorithm comparison is the size of the LCC divided by the size of the initial network, denoted as R_{gc} . The second step focuses on treating network decomposition as a whole process. It leads us to examine the dependence of $R_{gc}(f)$ on the proportion of removed edges, $f \in [0, 1]$, which represents the

decomposition degree independent of the size and topology of the initial network. Following the integral criterion proposed by Lubashevskiy et al. [29], [30], we introduce the area S_{rgc} under the curve $R_{gc}(f)$ for $0 \leq f \leq 1$ as a quantitative measure of decomposition efficiency. A smaller value of S_{rgc} indicates a faster decrease in LCC size and a more efficient algorithm in removing significant edges.

III. ITERATIVE APPROACH AND BENCHMARK RANKING INDICES

A. STATIC APPROACH

The Static approach to network decomposition proceeds as follows. Using an algorithm A , each edge E of an initial network is assigned a ranking value $\mathcal{A}$ that quantifies its significance. The edges are then ordered by their rankings:

$$E_{A,1}, E_{A,2}, E_{A,3}, \dots, E_{A,i}, \dots, E_{A,N} \quad (1)$$

such that for any pair $\{E_{A,i}, E_{A,j}\}$, the ranking indices satisfy the inequality $\mathcal{A}_i \geq \mathcal{A}_j$ for $i < j$, which is denoted as $E_{S,i} \geq E_{A,j}$. Here, N is the number of edges in the given network. Leaping ahead, we note that this ordering can exhibit potential degeneracy in cases where $\mathcal{A}_i = \mathcal{A}_j$ (for $i \neq j$) or when the difference $\mathcal{A}_i - \mathcal{A}_j \ll \mathcal{A}_i$ (for $i < j$) arises due to internal random factors in the algorithm's implementation. The function $R_{gc}(f)$ is constructed by removing the edges from the network according to sequence (1).

The main drawback of the Static approach is that the ordering (1) is sensitive to the network topology and can change significantly during the decomposition process. The essence of the Iterative approach is to reassess the significance of the remaining edges and reorder them after each edge removal.

B. ITERATIVE APPROACH

In the present paper, we analyze a number of widely used algorithms, ranking indices (RIs), for quantifying edge significance for several benchmark networks. The implementation of the Iterative approach for each algorithm is described by the following pseudocode for a selected RI and a network $\mathcal{N}$.

Step 0: Create an empty list L_E to be filled with the edges forming the final decomposition sequence, set the decomposition degree $f = 0$, the value of $R_{gc}(f = 0) = 1$, and $S_{rgc} = 0$

Step 1: Using the selected ranking index (RI), analyze the topological features of the network $\mathcal{N}$ at its *current state* $\mathcal{N}_f$ ($\mathcal{N}_{f=0} = \mathcal{N}$) and assign a ranking value $\mathcal{A}_{RI,i}$ to each edge $E_i \in \mathcal{N}_f$.

Step 2: Order the edges of the network $\mathcal{N}_f$ according to their ranking values $\{\mathcal{A}_{RI,i}\}_{i=1}^{i=N_f}$:

$$E_{RI,1}, E_{RI,2}, E_{RI,3}, \dots, E_{RI,i}, \dots, E_{RI,N_f} \quad (2)$$

such that $E_{RI,i} \geq E_{RI,j}$, i.e., $\mathcal{A}_{RI,i} \geq \mathcal{A}_{RI,j}$, for $i < j$.

Step 3: Remove the edge $E_{RI,1}$ from the network $\mathcal{N}_f$ and append it to the list L_E . The edge removal results in the transformations $f \rightarrow f + 1/N$ and $\mathcal{N}_f \rightarrow \mathcal{N}_f \setminus E_{RI,1}$.

Step 4: Find the largest connected component of the network $\mathcal{N}_f$. The corresponding value R_{gc} determines the magnitude of the function $R_{gc}(f)$ for the current decomposition degree f and the contribution $(1/N)R_{gc}$ to the desired criterion S_{rgc} of the IR efficiency for the given network $\mathcal{N}$.

Step 5: If the network $\mathcal{N}_f$ still contains edges, go to Step 1.

Step 6: When the network $\mathcal{N}_f$ contains no remaining edges, the list L_E represents the decomposition sequence of the original network $\mathcal{N}$, the dependence $R_{gc}(f)$ is constructed, and the desired criterion S_{rgc} is calculated.

Comparing the Iterative and Static approaches, it is important to emphasize that the obtained list

$$L_E = \{E_{RI,1}^L, E_{RI,2}^L, E_{RI,3}^L, \dots, E_{RI,i}^L, \dots, E_{RI,N}^L\} \quad (3)$$

is designed such that the edge $E_{RI,i}^L$ has the highest significance ranking, given that all preceding edges $\{E_{RI,j}^L\}_{j=0}^{i-1}$ have already been removed. In contrast, within the Static approach, the edge $E_{RI,i}$ is ranked as the i^{th} -most important only while all preceding edges remain in the network. Naturally, it is even more desirable to account not only for previously removed edges but also for the pattern of future disconnections. Achieving this, however, is feasible only within the Evolutionary approach to network decomposition [30].

C. BENCHMARK RANKING INDICES

Below, we outline the ranking indices (e.g., [23]) that will be analyzed for their efficiency in network decomposition.

Edge Betweenness Centrality [35] quantifies the significance of an edge by averaging the number of shortest paths between node pairs that pass through it. Specifically, the ranking value assigned to an edge E is

$$EB_E = \sum_{a \neq b} \frac{\sigma_{ab}(E)}{\sigma_{ab}}, \quad (4)$$

where $\sigma_{ab}(E)$ is the number of shortest paths between nodes a and b that pass through the edge E , and σ_{ab} is the total number of shortest paths between these nodes. The sum runs over all pairs of nodes $a \neq b$. Index (4) reflects global topology, as it accounts for the interconnections between all nodes in the network. Consequently, it can be highly sensitive to changes in the network structure.

Degree Product [36] is directly determined by the properties of the nodes incident to a given edge. Specifically, the ranking value assigned to an edge E is defined as:

$$DP_E = k_a \cdot k_b, \quad (5)$$

where k_a and k_b are the degrees of nodes a and b incident to the edge E . Index (5) is local in nature and is expected to be insensitive to network decomposition.

Bridgeness [37] is determined by the cliques associated with a given edge E and its adjacent nodes a and b . It is defined as:

$$B_E = \frac{\sqrt{S_a \cdot S_b}}{S_E}, \quad (6)$$

where S_a and S_b are the sizes of the largest cliques containing the nodes a and b , respectively, but not including the edge E . The value S_E represents the size of the largest clique to which the edge E belongs. Index (6) can be considered a semi-local or communal metric. It depends on the features of the nodes associated with the edge as well as the topological properties of the local cliques. However, modifications to distant parts of the network are unlikely to affect the Bridgeness value.

Diffusion Importance [38] focuses on information or influence spreading, where the intensity strongly depends on edge properties quantified by the corresponding edge significance [23], [44]. This index is defined as:

$$DI_E = \frac{n_{a \rightarrow b} + n_{b \rightarrow a}}{2}, \quad (7)$$

where a and b are the adjacent nodes of the analyzed edge E and $n_{a \rightarrow b}$ ($n_{b \rightarrow a}$) represents the numbers of links from the node $b(a)$ to nodes outside the nearest neighborhood of the node $a(b)$. Index (7) can be classified as a communal metric since it depends solely on the local network structure surrounding the edge E .

Topological Overlap, originally introduced to quantify tie strength in mobile communication networks [13], depends directly, like the Degree Product, on the degree properties of the nodes a and b adjacent to a given edge E . Specifically, it is defined as:

$$TO_E = \frac{n_{ab}}{k_a + k_b - n_{ab} - 2}, \quad (8)$$

where n_{ab} is number of nodes that are neighbors of both a and b , and, as before, k_a and k_b denote the degrees of nodes a and b , respectively.

Link Entropy [23], classified as a mid-level index from the network perspective, is constructed through two steps. First, the network nodes are partitioned into distinct communities using non-negative matrix factorization [45]. Second, the resulting matrix $\mathbf{X}$ is normalized and interpreted as a probabilistic clustering partition, where each element x_{ak} represents the probability of node a belonging to the community k . The Link Entropy index for an edge E , with adjacent nodes a and b , is defined as:

$$LE_E = \frac{H(\mathbf{X}_a) + H(\mathbf{X}_b) + 2 \cdot JSD(\mathbf{X}_a || \mathbf{X}_b)}{4}, \quad (9)$$

where $H(\mathbf{X}_a)$ denotes the information entropy, and $JSD(\mathbf{X}_a || \mathbf{X}_b)$ is the Jensen–Shannon divergence (for details

see [23]). When the number of assumed communities is relatively large, the LE_E -index for a given edge E can become insensitive to topological changes in distant communities. In such cases, the Link Entropy index may be regarded as a semi-local metric. Conversely, if the number of communities is small, any topological change can significantly impact the probabilistic clustering partition $\mathbf{X}$, rendering the LE_E -metric semi-global.

The *Link Entropy* algorithm involves two hidden parameters. One of them is the assumed number of communities, K , which serves as the basis for the non-negative matrix factorization. The other is the number of iterations, M , used during the implementation of this factorization. Following the analysis in [23], [26], [29], and [30], we set $K = 2$ and $M = 1000$ in our simulations to ensure the reliability of the obtained results.

k -Path Edge Centrality [39], [40] focuses on information propagation through simple paths consisting of at most k edges. For a given edge E , it is defined as:

$$k\text{-Path}_E = \sum_a \frac{\sigma_a^k(E)}{\sigma_a^k}, \quad (10)$$

where the sum runs over all the network nodes $\{a\}$, $\sigma_a^k(E)$ is the number of simple k -paths originating from node a and passing through the edge E within at most k -steps and, σ_a^k is the total number of simple k -paths originating from node a .

The k -Path Edge Centrality is similar to the Edge Betweenness but differs in two key aspects. First, while the Edge Betweenness considers paths of any length, the k -Path Edge Centrality restricts the length of paths to at most k edges. Second, the k -Path Edge Centrality accounts for all simple paths (excluding loops), not just the shortest ones. The parameter k determines the extent to which distant topological features influence the ranking index, making this metric mid-level for $1 \ll k \ll N$, where N is the size of the initial network. The implementation of Exp. (10) used here is based on the random walk formalism described in [40]. Referring to the findings in [46], $k = 20$ was chosen to ensure robust analysis.

IV. ANALYZED NETWORKS AND THEIR DECOMPOSITION

To facilitate a detailed comparison of the Static and Iterative approaches to network decomposition, we examined the corresponding destruction processes (i) using the seven ranking algorithms outlined in Sec. III-C, and (ii) across sixteen benchmark networks differing in size, topology, and the nature of represented systems. These networks encompass diverse domains, including social communication, physical infrastructures, biological connections, and human proximity, as detailed in the following subsection.

A. BENCHMARK NETWORKS

The selected networks and their descriptions were obtained from the Index of Complex Networks (ICON) project (<https://icon.colorado.edu/>) and the Netzschleuder catalog

TABLE 1. The benchmark networks used in the decomposition analysis. Their characteristic features shown in the table are the number of edges (Edges), the number of nodes (Nodes), the average node degree ($\langle k \rangle$), and the clustering coefficient (C).

Network \ Feature	Edges	Nodes	$\langle k \rangle$	C
Karate	77	34	4.53	0.26
Dolphins	159	62	5.13	0.31
Miserables	254	77	6.60	0.50
Transport	441	369	2.39	0.05
Football	613	115	10.66	0.41
Baseball	1,089	72	30.25	1.00
Gap Junction	1,450	469	6.18	0.19
Crime	1,476	1,380	2.14	0.00
Neurons	2,359	297	7.94	0.20
Jazz	2,742	198	27.70	0.52
Map of Science	6,523	687	18.99	0.47
Power	6,594	4,941	2.67	0.10
Workplace	9,827	92	213.63	0.46
Email	10,903	1,133	9.62	0.16
Proximities	28,244	274	206.16	0.90
Manufacturing	82,927	167	496.57	0.68

and repository of network datasets (<https://networks.skewed.de/>) as of September 23, 2024. These networks, along with their characteristic features, are presented in Table 1, arranged in ascending order of edge number, and are accompanied by a detailed description and reference links in the following list.

- **Karate:** A network of friendships among members of a university karate club [47].
- **Dolphins:** An undirected social network of frequent associations observed among 62 dolphins (*Tursiops*) in a community living off Doubtful Sound, New Zealand, from 1994 to 2001 [48].
- **Miserables:** The network of scene co-appearances of characters in Victor Hugo's novel *Les Misérables* [49].
- **Transport:** A multiplex network with three edge types representing links within the three layers of London train stations: Underground, Overground, and DLR [50].
- **Football:** A network of American football games between Division IA colleges during the regular season in Fall 2000 [35].
- **Baseball:** A network representing steroid use among baseball players, modeled as a bipartite network of players and their steroid providers (suppliers of illegal performance-enhancing substances) [51].
- **Gap Junction:** An undirected network among neurons of adult male and adult hermaphrodite *Caenorhabditis elegans* worms, constructed from electron microscopy series [52].
- **Crime:** A network of associations among suspects, victims, and/or witnesses involved in crimes in St. Louis in the 1990s, derived from police records via snowball sampling from five initial homicides [53].
- **Neurons:** A network representing the neural connections of the *Caenorhabditis elegans* nematode [54].
- **Jazz:** A network of collaborations among jazz musicians and jazz bands, extracted from *The Red Hot Jazz Archive*

digital database, covering bands that performed between 1912 and 1940 [55].

- **Map of Science:** A network of scientific fields extracted from English Wikipedia in early 2020, where nodes represent Wikipedia pages on natural, formal, social, and applied sciences, and two nodes are linked if the cosine similarity of the page content exceeds a threshold [56].
- **Power:** A network representing the Western States Power Grid of the United States, where nodes are transformers or power relay points, and two nodes are connected if a power line runs between them [57].
- **Workplace:** A temporal network of contacts between individuals, measured in an office building in France from June 24 to July 3, 2013 [58].
- **Email:** A network representing the exchange of emails among members of Rovira i Virgili University in Spain, in 2003 [59].
- **Proximities:** A network of human proximities measured using carried wireless devices. Each node represents a person, and an edge denotes when two individuals were within a certain proximity of each other, as detected by wireless protocols (Bluetooth or WiFi) [60].
- **Manufacturing:** A network of emails among employees at a mid-sized manufacturing company [61].

B. RESULTS

The results of the decomposition dynamics of the benchmark networks, described in Sec. IV-A, governed by the algorithms outlined in Sec. III-C, are summarized in Figs. 1–4 and Table 2.

Figures 1–4 illustrate the decomposition process studied numerically within the Static approach (Sec. III-A) and the Iterative approach (Sec. III-B). First, the left and middle columns in each figure depict the dependence of the Relative Size of the Largest Connected Component, $R_{gc}(f)$, on the decomposition degree f (proportion of removed edges), for the Static and Iterative approaches, respectively. Second, the right column displays bar plots of the found area S_{rgc} under the $R_{gc}(f)$ -curve, which serves as the primary quantitative measure of decomposition efficiency.

The bar plots compare the values of S_{rgc} obtained for seven ranking algorithms: *Edge Betweenness*, *Degree Product*, *Bridgeness*, *Diffusion Importance*, *Topological Overlap*, *Link Entropy*, and *k-Path Edge Centrality*. Blue and orange bars, corresponding to the Static and Iterative approaches, respectively, are grouped in pairs for easier comparison of algorithm efficiency between the two approaches. The same colors are used for the $R_{gc}(f)$ curves (left and middle columns) and the bar plot labels to indicate the corresponding ranking algorithms: *green* for *Edge Betweenness*, *red* for *Degree Product*, *violet* for *Bridgeness*, *light brown* for *Diffusion Importance*, *gray* for *Topological Overlap*, *yellow* for *Link Entropy*, and *light blue* for *k-Path Edge Centrality*. Finally, the rows in Figures 1–4 correspond to the results for

individual networks, arranged in ascending order of edge number, as listed in Table 1.

Table 2 presents the results in an alternative format. Each column corresponds to either the static or iterative version of the ranking algorithm used to assess edge significance. In the upper part of the table, each row represents a specific network, with the corresponding cells displaying the S_{rgc} -values obtained for that network when its decomposition is governed by the respective static or iterative ranking algorithm. In this way, the upper part of the table consolidates the data shown in the right columns of Figs. 1–4, but with greater numerical precision.

The lower part of this table presents a statistical analysis of the obtained results. Initially, we expected that, for all networks, ranking algorithms would yield better results when implemented within the Iterative approach compared to the Static approach. In other words, the value $S_{rgc}^{It}(\mathcal{N}, A)$, representing the decomposition of a given network $\mathcal{N}$ governed by the iterative version of a given ranking algorithm A , was expected to be smaller than $S_{rgc}^{St}(\mathcal{N}, A)$, which corresponds to the Static approach. However, the conducted investigation did not support this hypothesis in the general case. Therefore, to distinguish cases where the Iterative approach fails to improve network decomposition efficiency due to the specific features of a particular network from those where this outcome arises from inherent limitations of the ranking algorithm used, we conducted a statistical analysis of the results obtained for all seven benchmark algorithms. This analysis is presented in the rows of the table's lower part.

In this analysis, the null *right-tail* hypothesis posits that if the static version of a given ranking algorithm A has no inherent limitations and is perfectly efficient, then its corresponding iterative version cannot exhibit greater efficiency. In other words, the mean value

$$\bar{d}(A) \stackrel{\text{def}}{=} \frac{1}{16} \sum_{\mathcal{N}} \left[S_{rgc}^{St}(\mathcal{N}, A) - S_{rgc}^{It}(\mathcal{N}, A) \right] \quad (11)$$

must be equal to zero, $\bar{d}(A) = 0$. Here, the sum runs over all sixteen networks. It should be noted that this null right-tail hypothesis can also be interpreted as a hull hypothesis asserting that $\bar{d}(A) \leq 0$. The computed values of $\bar{d}$ are presented in the first row of the table's lower section. The second row displays the corresponding sample standard deviations, $s_d(A)$, calculated using the expression

$$s_d^2(A) = \frac{1}{15} \sum_{\mathcal{N}} \left[S_{rgc}^{St}(\mathcal{N}, A) - S_{rgc}^{It}(\mathcal{N}, A) - \bar{d}(A) \right]^2. \quad (12)$$

The third and forth row exhibit the corresponding t -values and p -values (calculated using Student's t distribution with degrees of freedom $\nu = 15$):

$$t\text{-value}(A) = \frac{\bar{d}(A)}{s_d(A)/\sqrt{16}}, \quad (13)$$

$$p\text{-value}(A) = t_{\nu}(t\text{-value}(A)). \quad (14)$$

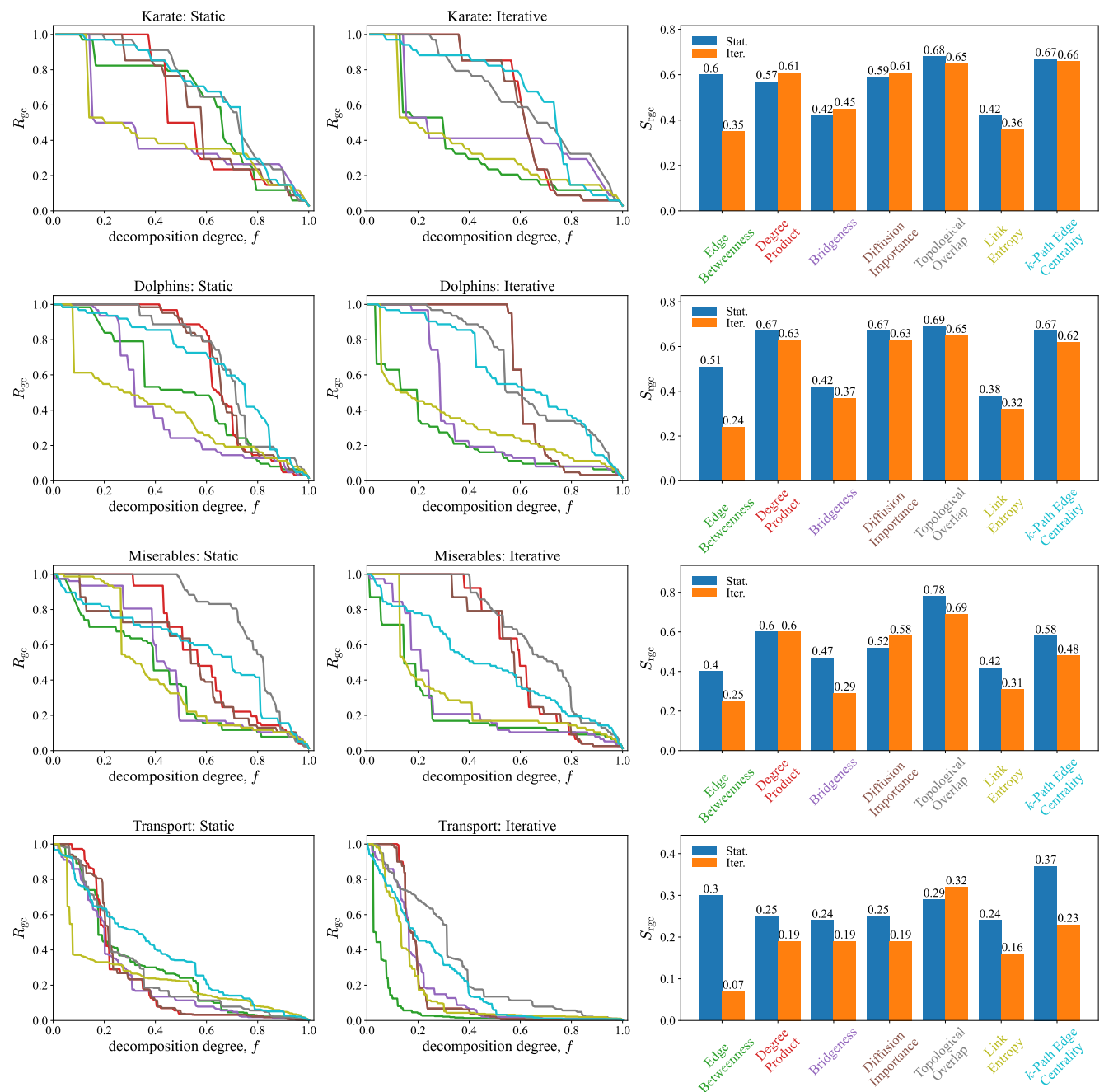


FIGURE 1. The decomposition process for four networks: Karate, Dolphins, Miserables, and Transport. Each row presents data for a specific network. The left column shows the decline in the Relative Size of the Largest Connected Component, R_{gc} , as network edges are removed in the order determined by the Static Edge Betweenness, Degree Product, Bridgeness, Diffusion Importance, Topological Overlap, Link Entropy, and k -Path Edge Centrality. The middle column illustrates the same $R_{gc}(f)$ -dynamics, obtained using the Iterative approach. The right column displays bar plots of the corresponding S_{gc} -values (areas under the $R_{gc}(f)$ -curves), which quantify the decomposition efficiency for the network destruction processes shown. The correspondence between specific $R_{gc}(f)$ -curves and the applied ranking indices is indicated by matching colors for the curves and bar plot labels.

The smaller the p -value, the stronger the evidence for an improvement in algorithm efficiency through the Iterative approach.

V. DISCUSSION

In discussing the obtained results, we focus on several key issues.

First, consistent with the initial expectation, the Iterative approach generally enhances the efficiency of network decomposition governed by the following ranking algorithms: Edge Betweenness, Bridgeness, Topological Overlap, Link Entropy, and k -path Edge Centrality. These algorithms can be categorized as ranking indices with a high- or mid-level scope of network integrity, meaning that they

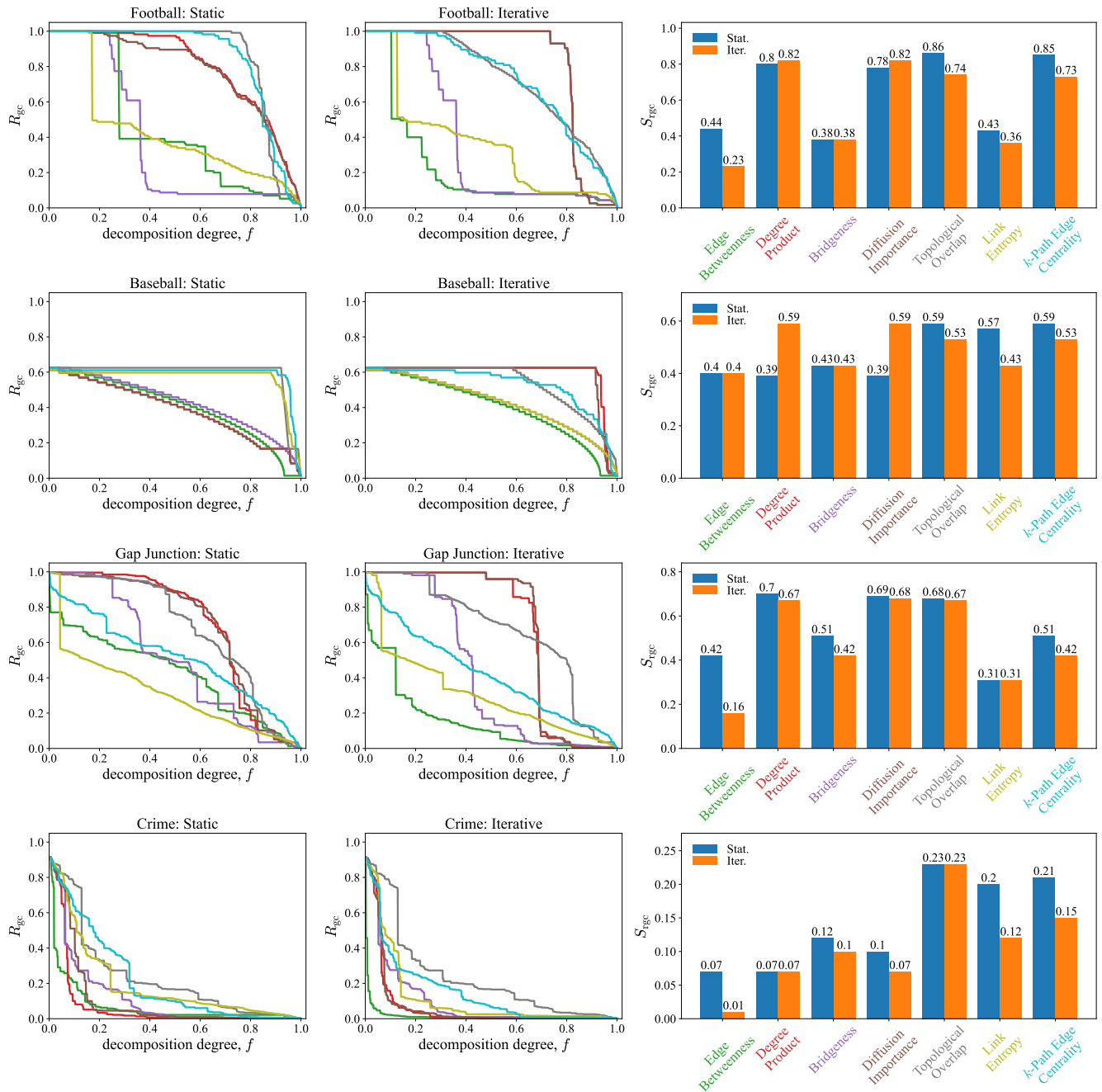


FIGURE 2. The decomposition process for four networks: Football, Baseball, Gap Junction, and Crime. Each row presents data for a specific network. The left column shows the decline in the Relative Size of the Largest Connected Component, R_{gc} , as network edges are removed in the order determined by the Static *Edge Betweenness*, *Degree Product*, *Bridgeness*, *Diffusion Importance*, *Topological Overlap*, *Link Entropy*, and *k-Path Edge Centrality*. The middle column illustrates the same $R_{gc}(f)$ -dynamics, obtained using the Iterative approach. The right column displays bar plots of the corresponding S_{rgc} -values (areas under the $R_{gc}(f)$ -curves), which quantify the decomposition efficiency for the network destruction processes shown. The correspondence between specific $R_{gc}(f)$ -curves and the applied ranking indices is indicated by matching colors for the curves and bar plot labels.

account for network fragments distant from the edge in question when quantifying edge significance. The efficiency enhancement, quantified by the ratio $\eta = S_{rgc}^{St}/S_{rgc}^{It} - 1$, warrants particular attention. When $\eta \ll 1$ (i.e., the efficiency improvement is minimal, only a few percent), the use of the Iterative version of the corresponding ranking algorithm

may not be well justified, as its computational cost increases by a factor of N_E , where N_E is the total number of edges. However, the situation changes when η becomes significant, $\eta \gtrsim 1$ or even $\eta \gg 1$, as observed, for example, with the *Edge Betweenness* algorithm applied to the *Transport*, *Crime*, *Map of Science*, and *Power* networks. Under such

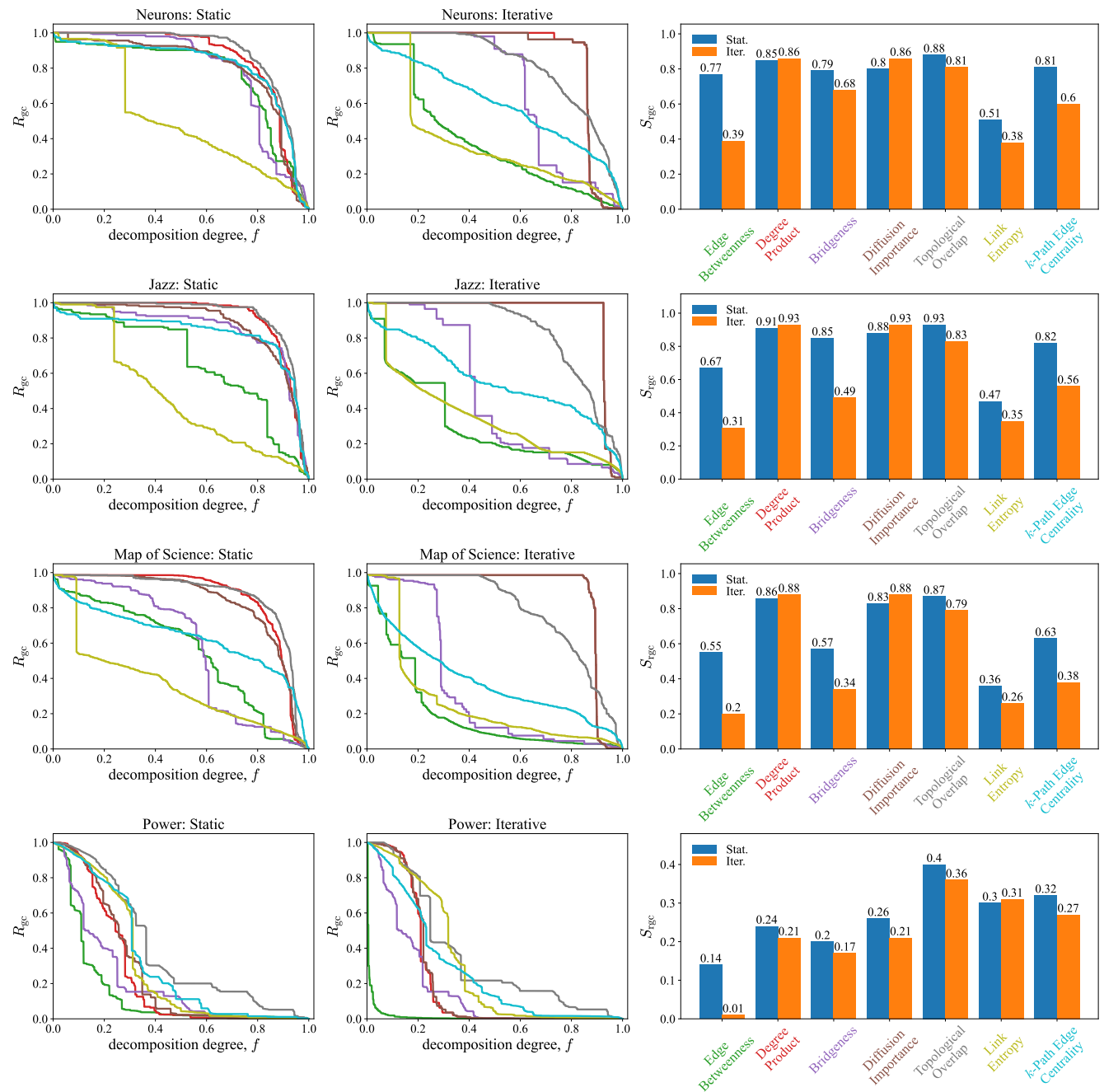


FIGURE 3. The decomposition process for four networks: Neurons, Jazz, Map of Science, and Power. Each row presents data for a specific network. The left column shows the decline in the Relative Size of the Largest Connected Component, R_{gc} , as network edges are removed in the order determined by the Static Edge Betweenness, Degree Product, Bridgeness, Diffusion Importance, Topological Overlap, Link Entropy, and k -Path Edge Centrality. The middle column illustrates the same $R_{gc}(f)$ -dynamics, obtained using the Iterative approach. The right column displays bar plots of the corresponding S_{gc} -values (areas under the $R_{gc}(f)$ -curves), which quantify the decomposition efficiency for the network destruction processes shown. The correspondence between specific $R_{gc}(f)$ -curves and the applied ranking indices is indicated by matching colors for the curves and bar plot labels.

conditions, the Iterative approach provides a clear advantage in decomposition dynamics. In this context, analyzing other ranking indices for their efficiency within the Iterative approach could be important for both practical and academic purposes, particularly when achieving efficient network decomposition outweighs concerns about computational cost.

Second, as an unexpected result, the analysis revealed no statistically significant evidence that the Iterative approach enhances the decomposition efficiency in the case of *Degree Product* and *Diffusion Importance*. However, this does not justify dismissing these ranking indices as unsuitable for network decomposition. Both indices have a long history

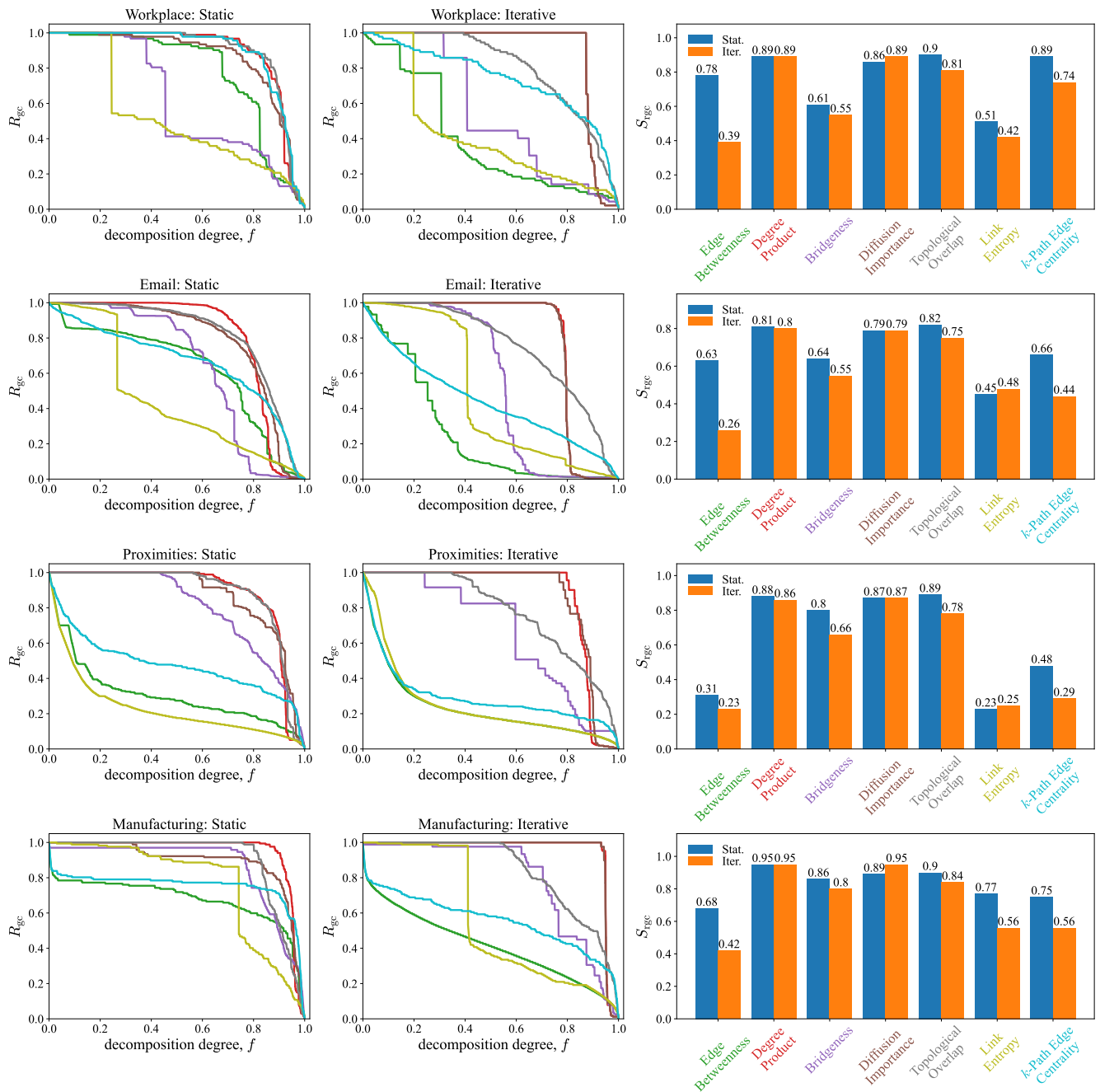


FIGURE 4. The decomposition process for four networks: Workplace, Email, Proximities, Manufacturing. Each row presents data for a specific network. The left column shows the decline in the Relative Size of the Largest Connected Component, R_{gc} , as network edges are removed in the order determined by the Static Edge Betweenness, Degree Product, Bridgeness, Diffusion Importance, Topological Overlap, Link Entropy, and k -Path Edge Centrality. The middle column illustrates the same $R_{gc}(f)$ -dynamics, obtained using the Iterative approach. The right column displays bar plots of the corresponding S_{gc} -values (areas under the $R_{gc}(f)$ -curves), which quantify the decomposition efficiency for the network destruction processes shown. The correspondence between specific $R_{gc}(f)$ -curves and the applied ranking indices is indicated by matching colors for the curves and bar plot labels.

of practical and academic use, and for certain networks, such as *Crime*, they demonstrated efficiency superior to that of other methods. In fact, the results suggest that the Iterative approach reveals some latent limitations—hidden insufficiency—of these ranking indices, which operate with a low-level scope of network integrity. These limitations

may stem from the presence of numerous edges with similar ranking values, where removing the first edge encountered, rather than all critical ones, fails to substantially enhance network decomposition. Other ranking algorithms also exhibited latent inefficiencies. For example, *Link Entropy* showed a degradation in efficiency for the *Power*,

TABLE 2. Table of the numerical data on methods efficiency (S_{rgc} -values) in its Static and Iterative forms. The bottom part of the table represents statistical data on the difference in the efficiency of Static and Iterative forms, and the p-values correspond to the type one error for concluding that the iterative approach improves the efficiency of a particular method.

Network \ Algorithm	Edge Betweenness		Degree Product		Bridgeness		Diffusion Importance		Topological Overlap		Link Entropy		k -Path Edge Centrality	
	Stat.	Iter.	Stat.	Iter.	Stat.	Iter.	Stat.	Iter.	Stat.	Iter.	Stat.	Iter.	Stat.	Iter.
Karate	0.6029	0.3488	0.5683	0.6146	0.4208	0.4521	0.5890	0.6101	0.6791	0.6497	0.4212	0.3624	0.6678	0.6580
Dolphins	0.5110	0.2437	0.6685	0.6296	0.4218	0.3678	0.6741	0.6296	0.6946	0.6508	0.3819	0.3158	0.6736	0.6219
Miserables	0.3969	0.2478	0.6024	0.6018	0.4664	0.2899	0.5177	0.5849	0.7759	0.6861	0.4248	0.3107	0.5774	0.4810
Transport	0.2974	0.0660	0.2484	0.1936	0.2442	0.1904	0.2522	0.1925	0.2877	0.3154	0.2388	0.1648	0.3650	0.2309
Football	0.4444	0.2294	0.8041	0.8240	0.3785	0.3777	0.7816	0.8240	0.8595	0.7353	0.4264	0.3619	0.8503	0.7317
Baseball	0.4016	0.4021	0.3890	0.5933	0.4305	0.4305	0.3890	0.5857	0.5946	0.5339	0.5690	0.4296	0.5913	0.5301
Gap Junction	0.4168	0.1556	0.6960	0.6712	0.5068	0.4207	0.6931	0.6803	0.6757	0.6669	0.3073	0.3064	0.5114	0.4176
Crime	0.0712	0.0142	0.0705	0.0679	0.1193	0.0997	0.1036	0.0687	0.2348	0.2348	0.1955	0.1156	0.2137	0.1489
Neurons	0.7706	0.3854	0.8521	0.8639	0.7895	0.6799	0.8034	0.8610	0.8841	0.8141	0.5077	0.3849	0.8115	0.5971
Jazz	0.6660	0.3115	0.9136	0.9327	0.8536	0.4876	0.8770	0.9327	0.9251	0.8338	0.4730	0.3540	0.8233	0.5600
Map of Science	0.5470	0.1991	0.8557	0.8804	0.5709	0.3438	0.8293	0.8804	0.8670	0.7858	0.3590	0.2643	0.6301	0.3841
Power	0.1415	0.0133	0.2409	0.2102	0.2028	0.1696	0.2634	0.2144	0.3960	0.3550	0.2956	0.3061	0.3216	0.2729
Workplace	0.7794	0.3946	0.8937	0.8901	0.6101	0.5472	0.8607	0.8901	0.8969	0.8050	0.5135	0.4177	0.8933	0.7390
Email	0.6251	0.2591	0.8089	0.7955	0.6408	0.5472	0.7916	0.7940	0.8176	0.7522	0.4506	0.4758	0.6621	0.4424
Proximities	0.3063	0.2329	0.8841	0.8639	0.7996	0.6640	0.8706	0.8660	0.8854	0.7750	0.2320	0.2474	0.4762	0.2939
Manufacturing	0.6754	0.4151	0.9481	0.9520	0.8648	0.7961	0.8940	0.9514	0.9039	0.8405	0.7680	0.5564	0.7522	0.5579
Characteristics		Algorithm Statistics												
$\bar{d} = \langle \text{Stat.} - \text{Iter.} \rangle$		0.2334	-0.0088		0.0910		-0.0235		0.0590		0.0744		0.1346	
$s_d = \sqrt{\langle (d - \bar{d})^2 \rangle}$		0.1223	0.0582		0.0992		0.0628		0.0413		0.0637		0.0784	
t -value		7.6344	-0.6029		3.6692		-1.4946		5.7100		4.6719		6.8621	
p -value		7.623e-07	0.7222		0.0011		0.9221		2.065e-05		1.505e-04		2.697e-06	

Email, and *Proximities* networks. Investigating whether this degradation is related to a limited number of iterations during community construction or to incorrect community identification [29] merits further exploration. Today, a wide range of ranking indices has been proposed, including those already considered benchmark methods and those still under development. The Iterative approach offers a valuable tool for uncovering hidden features and potential issues in these indices, paving the way for a deeper understanding and further improvement.

Third, the results have demonstrated that different edge-ranking indices might be preferable for different networks, suggesting that investigating the dependence of ranking index performance on initial network features could be a subject of future research. Intuitively, if iterative re-assessment of edge significance using the same ranking index produces a more efficient edge decomposition sequence, then combining different ranking indices within a single decomposition process could be the next step toward a more comprehensive understanding of edge criticality assessment.

Fourth, the aim of this article is to bridge the gap between the classic static approach to edge criticality assessment for network decomposition and the recent Evolutionary approach (EA) [30]. A key drawback of the EA is the significant amount of computation time required to generate a decomposition plan from an initially random set of seeds. It has been hypothesized that computation time could be reduced by combining the Static approach and the EA for generating reasonable seeds based on conventional ranking indices and then employing the EA for further improvement of the decomposition plan. In this context, the Iterative approach offers better results than static methods while requiring less time than the EA. Furthermore, while the use of static ranking indices to generate initial seeds for the EA may

be reasonable for certain applications, using seeds generated by the Iterative approach is an even better option, as they are closer to the optimal solution.

Fifth, in the present work, we propose the Iterative approach to edge significance assessment that involves re-evaluating topological features after each individual edge removal. Such frequent recalculation of ranking indices for all edges naturally increases the time complexity of the process. To address this issue, the frequency of re-evaluation could be reduced, for instance, by performing it after every two, five, or ten-edge removals. However, to avoid bias, it is important to investigate how the optimal frequency of edge significance re-assessment depends on the network's topological features and the choice of ranking indices.

VI. CONCLUSION

Identification of critical edges for optimal network decomposition or, conversely, their protection is a challenging issue in network science. Currently, two main approaches are well recognized. The first is the Static approach, which involves applying an algorithm to assess edge-ranking indices and then removing edges one by one in the determined order. The second is the Optimization-based approach, where network decomposition is treated as an integral process, and the decomposition plan is optimized using genetic-like algorithms. Such algorithms operate on sequences of all edges, ordered by their removal from the given network, as fundamental elements. These elements are compared using integral criteria and undergo individual or pairwise transformations to accelerate algorithm execution. However, the approach remains highly time-consuming. The Static approach is the fastest to implement, while the Optimization-based approach has the potential to yield

superior results, as it considers the edge decomposition sequence as a whole.

In this work, we propose the Iterative approach to bridge the gap between the Static and Optimization-based approaches. The key idea is to use a ranking index to identify one or more of the most critical edges, remove them, and then re-evaluate edge significance for the modified network. By repeating this process until the network is completely disconnected, the Iterative approach achieves a better decomposition pattern than the Static approach while being significantly faster than the Optimization-based approach. To explain this, we note that although the Iterative approach re-evaluates and reorders the remaining edges after each removal, it does not compare the newly generated edge sequence with multiple possible sequences using an integral criterion, nor does it combine them to achieve a more optimal decomposition pattern. As noted above, such comparison and combination are intrinsic to the Optimization-based approach. Therefore, the Iterative approach is expected to be faster, while the Optimization-based approach should yield a superior decomposition.

To evaluate the Iterative approach, we conducted numerical analyses on the decomposition of sixteen benchmark networks using seven widely adopted ranking algorithms within both the Static and Iterative frameworks: *Edge Betweenness*, *Degree Product*, *Bridgeness*, *Diffusion Importance*, *Topological Overlap*, *Link Entropy*, and *k-Path Edge Centrality*. The results demonstrate, first, that the efficiency of network decomposition can increase tenfold when using the Iterative approach. Second, they reveal that certain ranking algorithms exhibit inherent insufficiency in network destruction, leading to reduced efficiency within the Iterative approach for some networks. This finding underscores the need for a deeper understanding of the mechanisms underlying network decomposition and paves the way for verifying the completeness of ranking algorithms or identifying and addressing their insufficiency.

Many challenging issues remain in critical edge detection, reflecting various aspects of the problem, and their solutions could benefit from the proposed Iterative approach. In particular, employing the Iterative approach to generate decomposition patterns for the main benchmark networks using standard ranking indices could provide insights into the dependence of decomposition efficiency on network topology and the chosen ranking index. This analysis would enable the identification of the optimal ranking index.

Further improvements to the Iterative approach may involve:

- Balancing the frequency of edge criticality re-assessment, which would reduce the time complexity of the proposed method.
- Mixing ranking indices, switching between them during the decomposition process in response to topological changes in the network.

As a principal limitation of the Iterative approach, we note that it involves repeatedly applying the same edge-ranking procedure for the removal of high-ranked edges after each or a few steps of edge removal. However, during its implementation, the Iterative approach inherently creates new potentially effective decomposition patterns. This opens the possibility for improving the Iterative approach by incorporating elements of the Evolutionary approach, specifically by mixing previously created decomposition patterns with newly generated ones and comparing the results using integral criteria. On the one hand, the collection of possible decomposition patterns should not be very large, as it is created by turning to other optimal criteria. On the other hand, by combining these patterns, it could be possible to merge edge-ranking algorithms with the discovery of hidden optimal edge removal configurations specific to different networks.

Additionally, it would be worthwhile to apply the Iterative approach to the problem of critical node identification. Given that improving edge criticality rankings enhances efficiency, it is plausible that node criticality rankings would similarly improve.

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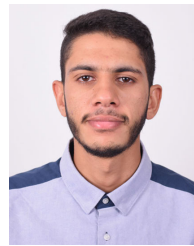
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